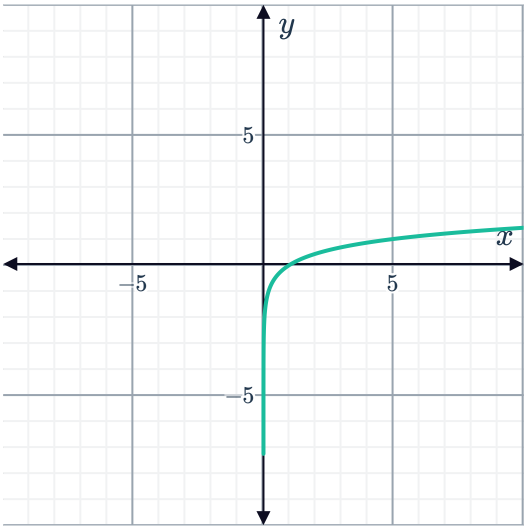
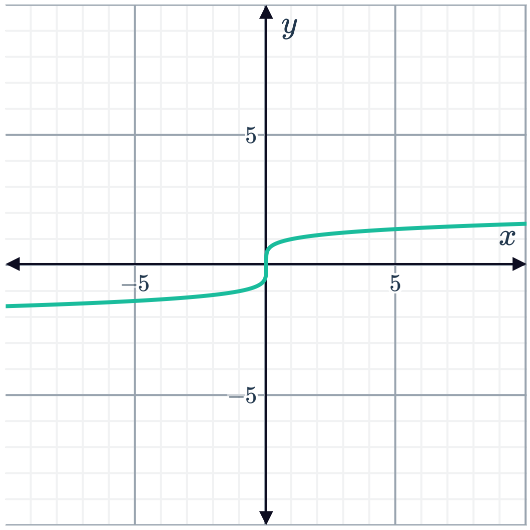
Graphs of logarithmic functions

1 Consider the functions graphed below:
Which of these graphs represents a logarithmic function of the form $y = \log_a(x)$?

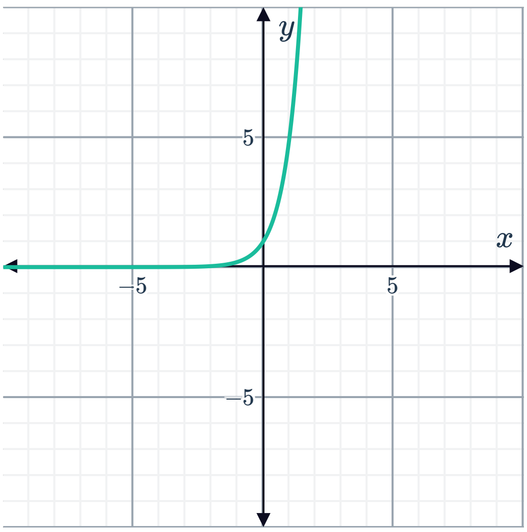
A



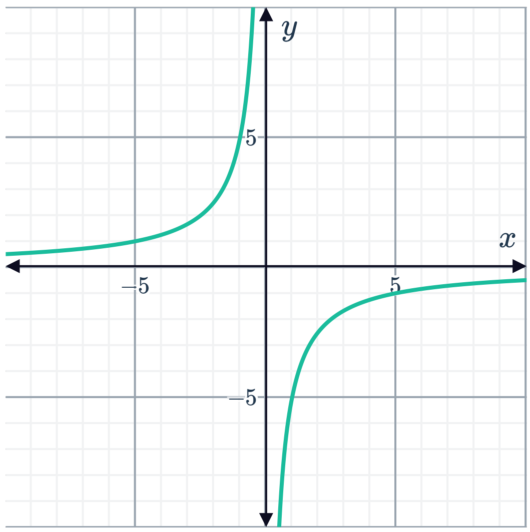
B



C



D



2 Consider the function $y = \log_2 x$.

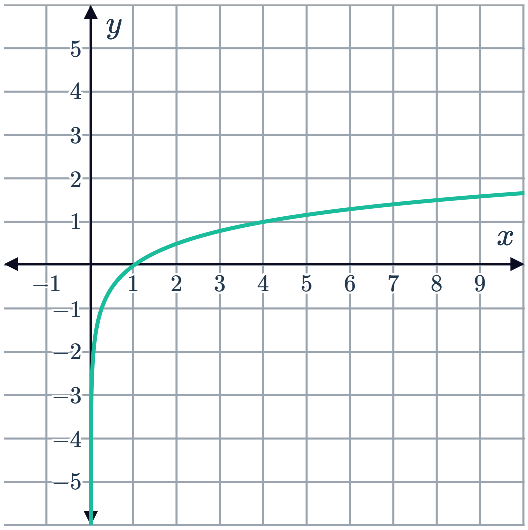
a Complete the following table of values:

x	$\frac{1}{2}$	1	2	4	16
y					

b Sketch a graph of the function.

c State the equation of the vertical asymptote.

3 Consider the function $y = \log_4 x$ and its given graph:



a Complete the following table of values:

x	$\frac{1}{16}$	$\frac{1}{4}$	4	16	256
y					

b Find the x-intercept.

c How many y-intercepts does the function have?

d Find the x-value for which $\log_4 x = 1$.

4 Sketch the graph of $y = \log_5 x$.

5 Consider the function $y = \log_4 x$.

a Complete the table of values.

x	$\frac{1}{1024}$	$\frac{1}{4}$	1	4	16	256
y						

b Is $\log_4 x$ an increasing or decreasing function?

c Describe the behaviour of $\log_4 x$ as x approaches 0.

d State the value of y when $x = 0$.

6 Consider the function $y = \log_a x$, where a is a value greater than 1.

a For which of the following values of x will $\log_a x$ be negative?

A $x = -9$ **B** $x = \frac{1}{9}$ **C** $x = 9$ **D** $\log_a x$ is never negative

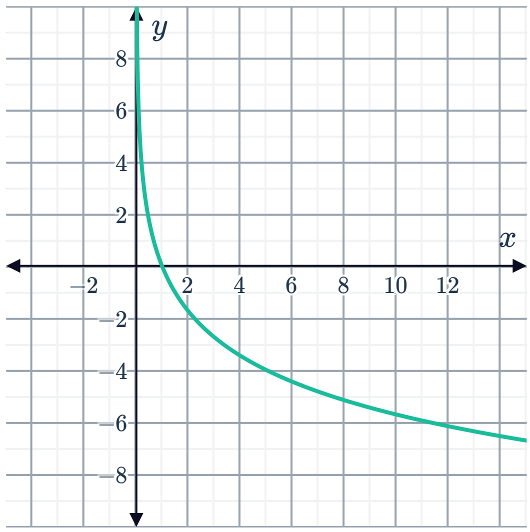
b For which of the following values of x will $\log_a x$ be positive?

A $x = 5$ **B** $x = -5$ **C** $x = \frac{1}{5}$ **D** $\log_a x$ will never be positive

c Is there a value that $\log_a x$ will always be greater than?

d Is there a value that $\log_a x$ will always be less than?

7 Consider the given graph of the logarithmic function $y = \log_a x$:



a Is $\log_a x$ an increasing or decreasing function?

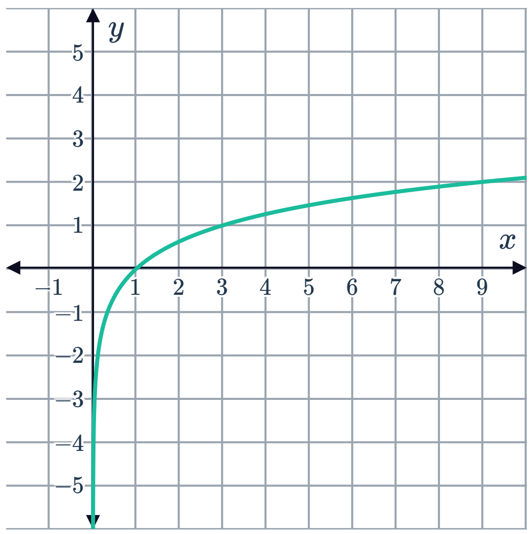
b Which is a possible value for a, $\dfrac{2}{3}$ or $\dfrac{3}{2}$?

8 Consider the functions $y = \log_2 x$ and $y = \log_3 x$.

a Sketch the two functions on the same set of axes.

b Describe how the size of the base relates to the steepness of the graph.

9 Consider the given graph of $f(x) = \log_k x$:



a Determine the value of the base k.

b Hence, state the equation of $f(x)$.

Transformations of logarithmic functions

10 Consider the functions $f(x) = \log_2 x$ and $g(x) = \log_2 x + 2$.

a Complete the table of values below:

x	$\frac{1}{2}$	1	2	4	8
$f(x) = \log_2 x$					
$g(x) = \log_2 x + 2$					

b Sketch the graphs of $y = f(x)$ and $y = g(x)$ on the same set of axes.

c Describe a transformation that can be used to obtain $g(x)$ from $f(x)$.

d Determine whether each of the following features of the graph will remain unchanged after the given transformation:

i The vertical asymptote.

ii The general shape of the graph.

iii The x-intercept.

iv The range.

11 Consider the functions $f(x) = \log_2(-x)$ and $g(x) = \log_2(-x) - 3$.

a Complete the table of values below:

x	-8	-4	-2	-1	$-\frac{1}{2}$
$f(x) = \log_2(-x)$					
$g(x) = \log_2(-x) - 3$					

b Sketch the graphs of $y = f(x)$ and $y = g(x)$ on the same set of axes.

c Describe a transformation that can be used to obtain $g(x)$ from $f(x)$.

d Determine whether each of the following features of the graph will remain unchanged after the given transformation:

i The vertical asymptote.

ii The general shape of the graph.

iii The x-intercept.

iv The domain.

12 Sketch the graph of the following functions:

a $y = \log_3 x$ translated 2 units up.

b $y = \log_3 x$ translated 4 units down.

c $y = \log_2 x + 4$.

13 The graph of $y = \log_6 x$ is transformed to c[illegible] the graph of $y = \log_6 x + 4$. Describe a tranformation that could achieve this.

14 For each of the following functions:

i State the equation of the function after it has been translated.

ii Sketch the translated graph.

a $y = \log_5 x$ translated downwards by 2 units.

b $y = \log_3 (-x)$ translated upwards by 2 units.